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Bidirectional Coverttness and Security Enhancement in FD-ISAC Systems

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Abstract—To mitigate diverse security threats in 6G full-duplex integrated sensing and communication (FD-ISAC) systems, this paper presents a bidirectional security enhancement framework. This framework simultaneously supports heterogeneous services involving uplink (UL) public and covert access, downlink (DL) secure transmission, and continuous radar sensing within a unified spectrum. Specifically, we exploit the intrinsic high-power radar and public user signals to mask UL covert transmissions while strategically embedding artificial noise into DL radar beams to degrade eavesdropping channels. We establish a joint beamforming and power optimization problem to maximize the aggregate covert and secrecy spectral efficiency. To address channel state information (CSI) uncertainty in the surveillance link, a worst-case constraint based on the S-procedure is derived to strictly guarantee coverttness. The resulting non-convex problem is efficiently solved via successive convex approximation (SCA) and rank relaxation. Numerical simulations demonstrate the effective covert communication capability and robust security performance achieved by our proposed scheme.

Index Terms—Integrated sensing and communication (ISAC), full-duplex (FD), coverttness and security, artificial noise (AN).

I. INTRODUCTION

Full-duplex (FD) technology has been widely adopted in integrated sensing and communication (ISAC) systems to enhance spectral efficiency (SE). However, with the anticipated proliferation of FD-ISAC systems, securing them against malicious attacks has become a paramount concern [1], [2], [3], [4], [5]. Specifically, the inherent vulnerability of sensing waveforms to eavesdropping necessitates physical layer security (PLS) [2]. While existing studies have extensively explored PLS techniques—such as beamforming optimization and artificial noise (AN) injection—to maximize secrecy rates [6], [7], they primarily protect the data content, leaving the transmission behavior itself exposed. Against sophisticated adversaries capable of side-channel analysis, data secrecy alone is insufficient; making the sheer existence of the communication invisible is imperative [8]. To this end, prior studies have investigated various aerial [9], [10], [11], [12] and jamming-aided [13], [14] techniques to actively conceal the transmission behavior.

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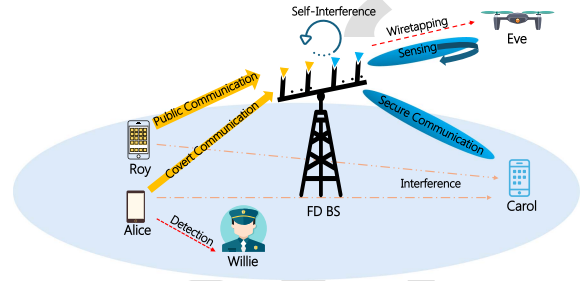


Fig. 1. System model.

To actively enhance communication security and coverttness, recent research has capitalized on the intrinsic dual-functional nature of ISAC by exploiting radar sensing signals. Specifically, the authors in [15] utilized an FD base station to generate sensing-assisted jamming, effectively safeguarding uplink (UL) transmissions against aerial eavesdroppers. Similarly, radar interference was repurposed in [14] to mask covert communications. While these single-objective strategies perform well in isolated scenarios, highly dynamic wireless networks frequently encounter coordinated adversaries capable of simultaneous detection and interception. Consequently, mitigating such multifaceted risks requires moving beyond isolated defenses to cohesively integrate PLS and coverttness. Addressing this need, the study in [16] leveraged a simultaneously transmitting and reflecting reconfigurable intelligent surface (STAR-RIS) to enable joint covert and secure downlink (DL) communications against both an eavesdropper and a warden. This framework was subsequently extended to multi-adversary ISAC systems in [17].

While existing works advance unidirectional security, they typically rely on auxiliary hardware (e.g., STAR-RIS) or active beamforming. As summarized in Table I, exploiting FD-ISAC's inherent dual-functional signals for bidirectional security remains largely under-explored. Besides, Willie's passive nature makes acquiring perfect surveillance channel state information (CSI) practically impossible; ignoring this uncertainty would immediately compromise theoretical coverttness.

Motivated by this, our framework exploits concurrent DL signals to naturally mask UL traffic. This achieves coverttness without burdening resource-constrained UL users, reserving active beamforming for DL security. The main contributions of this paper are summarized as follows: 1) Heterogeneous Services: We propose a unified framework that simultaneously supports UL public and covert communications alongside DL secure transmission, all while maintaining radar sensing; 2) Strategy: We exploit the inherent high-power radar waveforms and concurrent public signals to naturally mask covert traffic, obviating the need for external jamming; 3) Optimization: We maximize sum SE through joint beamforming and power optimization, developing a worst-case robust design to guarantee robustness against CSI uncertainty.

II. SIGNAL MODEL AND COVERT REQUIREMENT

As illustrated in Fig. 1, a full-duplex base station (FD-BS) equipped with N_t transmit and N_r receive antennas facilitates bidirectional ISAC

TABLE I
COMPARISON OF THIS WORK WITH EXISTING LITERATURE

Ref.	System Model	Objectives		Robustness	Key Enablers (Covert / Secure)
		PLS	Covert		
[2]	Bidirectional FD-ISAC	✓	-	×	- / Active Beamforming + AN
[6]	DL ISAC	✓	-	✓	- / Active Beamforming + AN
[14]	DL ISAC	-	✓	✓	Active Beamforming / -
[15]	UL FD-ISAC	✓	-	✓	- / Active Beamforming
[16]	DL mmWave	✓	✓	×	Overt Signals & Active Beamforming
[17]	DL ISAC	✓	✓	✓	Overt Signals & Active Beamforming
Ours	Bidirectional FD-ISAC	✓	✓	✓	Concurrent Radar & Overt Signals / Active Beamforming + AN

services under dual security threats. In the uplink, a warden (Willie) attempts to detect Alice's covert signal (s_a). Alice opportunistically conceals her transmission within the continuous public traffic of an aggregate user, Roy (s_r), whose natural communication inadvertently serves as ambient interference to confound Willie. Concurrently, in the downlink, the FD-BS transmits a dual-functional signal ($\mathbf{x}$) to securely communicate with Carol while tracking a target (e.g., an unauthorized drone, Eve) that doubles as an eavesdropper. Consequently, the FD-BS faces the conflicting task of adequately illuminating Eve for precise tracking while actively degrading her interception capability.

Let s_k ($k \in \{a, r\}$) denote the transmitted signals of Alice and Roy, with transmit powers bounded by $p_k = \mathbb{E}[|s_k|^2] \leq P_{\max}$. Concurrently in the downlink, the FD-BS transmits a composite signal $\mathbf{x} = \mathbf{v}_c s_c + \mathbf{s}_0$, where $s_c \sim \mathcal{CN}(0, 1)$ is the confidential symbol for Carol precoded by $\mathbf{v}_c$, and $\mathbf{s}_0 \sim \mathcal{CN}(\mathbf{0}, \mathbf{V}_0)$ is the artificial noise (AN) vector. The aggregate transmit covariance is thus $\mathbf{R}_{xx} = \mathbb{E}[\mathbf{x}\mathbf{x}^H] = \mathbf{V}_c + \mathbf{V}_0$, with $\mathbf{V}_c = \mathbf{v}_c \mathbf{v}_c^H$ and $\mathbf{V}_0 \succeq \mathbf{0}$.

Assuming independent Rayleigh fading and available channel state information (CSI) for all links, the uplink channel from user k to the FD-BS is modeled as $\mathbf{h}_{kb} \sim \mathcal{CN}(\mathbf{0}, \sigma_{kb}^2 \mathbf{I}_{N_r})$. Similarly, the surveillance links to Willie from user k and the FD-BS are denoted by $g_{kw} \sim \mathcal{CN}(0, \sigma_{kw}^2)$ and $\mathbf{g}_{bw} \sim \mathcal{CN}(\mathbf{0}, \sigma_{bw}^2 \mathbf{I}_{N_t})$, respectively. To detect Alice's covert behavior, Willie performs a binary hypothesis test, where his received signal under the null hypothesis $\mathcal{H}_0$ (Alice silent) and the alternative hypothesis $\mathcal{H}_1$ (Alice transmits) is formulated as:

$$y_W = \begin{cases} g_{rw} s_r + \mathbf{g}_{bw}^H \mathbf{x} + n_w, & \mathcal{H}_0 \\ g_{rw} s_r + \mathbf{g}_{bw}^H \mathbf{x} + g_{aw} s_a + n_w, & \mathcal{H}_1 \end{cases} \quad (1)$$

where $n_w \sim \mathcal{CN}(0, \sigma_w^2)$ denotes the background noise.

For the downlink (DL) transmissions, let $\mathbf{q}_{bc} \in \mathbb{C}^{N_t \times 1}$ denote the Rician fading channel to the legitimate user Carol. Since untrusted aerial targets typically act as malicious nodes [2], the wiretap channel to the eavesdropper (Eve) at angle θ_0 is dominated by a line-of-sight (LoS) component $\alpha_0 \mathbf{a}_t^H(\theta_0)$, where α_0 is the path-loss coefficient. The transmit steering vector is $\mathbf{a}_t(\theta) = \mathbf{a}_{N_t}(\theta)$, with $\mathbf{a}_{N_t}(\theta) = [1, e^{j\frac{2\pi}{\lambda} d \sin(\theta)}, \dots, e^{j\frac{2\pi}{\lambda} (N-1) d \sin(\theta)}]^T$. Let f_{lc} and u_{le} ($\ell \in \{r, a\}$) denote the UL-to-DL interference channels, modeled as independent Rayleigh fading $\mathcal{CN}(0, \sigma_\ell^2)$. The received signals at Carol and Eve are respectively formulated as

$$y_C = f_{rc} s_r + f_{ac} s_a + \mathbf{q}_{bc}^H \mathbf{x} + n_c, \quad (2)$$

$$y_E = u_{re} s_r + u_{ae} s_a + \alpha_0 \mathbf{a}_t^H(\theta_0) \mathbf{x} + n_p, \quad (3)$$

where $n_c \sim \mathcal{CN}(0, \sigma_c^2)$ and $n_p \sim \mathcal{CN}(0, \sigma_p^2)$ denote the respective additive noises, and the receive steering vector is $\mathbf{a}_r(\theta) = \mathbf{a}_{N_r}(\theta)$. Consequently, the DL SINRs for Carol and Eve are respectively formulated as

$$\Upsilon_c = \frac{\mathbf{q}_{bc}^H \mathbf{V}_c \mathbf{q}_{bc}}{p_r |f_{rc}|^2 + p_a |f_{ac}|^2 + \mathbf{q}_{bc}^H \mathbf{V}_0 \mathbf{q}_{bc} + \sigma_c^2}, \quad (4)$$

$$\Upsilon_e = \frac{|\alpha_0|^2 \mathbf{a}_t^H(\theta_0) \mathbf{V}_c \mathbf{a}_t(\theta_0)}{p_r |u_{re}|^2 + p_a |u_{ae}|^2 + |\alpha_0|^2 \mathbf{a}_t^H(\theta_0) \mathbf{V}_0 \mathbf{a}_t(\theta_0) + \sigma_p^2}. \quad (5)$$

Simultaneously, the target's radar echo is expressed as $\mathbf{y}_r = \beta_0 \mathbf{a}_r(\theta_0) \mathbf{a}_t^H(\theta_0) \mathbf{x} + \mathbf{c}$, where β_0 represents the complex reflection coefficient accounting for both two-way path loss and radar cross section (RCS). Unlike the one-way communication channel $\alpha_0 \mathbf{a}_t^H(\theta_0)$, the two-way radar echo inherently incorporates target-specific backscatter; thus, β_0 and α_0 are modeled independently despite their shared macroscopic geometry. The environmental clutter is characterized by $\mathbf{c} \sim \mathcal{CN}(\mathbf{0}, \mathbf{R}_c)$ with a constant covariance matrix $\mathbf{R}_c \in \mathbb{C}^{N_r \times N_r}$ [2]. Furthermore, FD operation introduces self-interference (SI), modeled as $\sqrt{\beta} \mathbf{H}_{SI} \in \mathbb{C}^{N_r \times N_t}$ with residual SI power β . Aggregating these components alongside the UL signals and background noise $\mathbf{n}_b \sim \mathcal{CN}(\mathbf{0}, \sigma_b^2 \mathbf{I}_{N_r})$, the overall received signal at the FD-BS is formulated as

$$\mathbf{y}_{BS} = \mathbf{h}_{rb} s_r + \mathbf{h}_{ab} s_a + \mathbf{y}_r + \sqrt{\beta} \mathbf{H}_{SI} \mathbf{x} + \mathbf{n}_b. \quad (6)$$

Assuming N independent channel uses, the transmitted signals and noise are modeled as independent Gaussian processes. Consequently, each received sample at Willie follows a circularly symmetric complex Gaussian distribution, i.e., $y_W[n] | \mathcal{H}_i \sim \mathcal{CN}(0, \lambda_i)$. The expected received power $\lambda_i \triangleq \mathbb{E}[|y_W[n]|^2 | \mathcal{H}_i]$ under each hypothesis is defined as

$$\begin{aligned} \lambda_0 &= p_r |g_{rw}|^2 + \mathbf{g}_{bw}^H \mathbf{R}_{xx} \mathbf{g}_{bw} + \sigma_w^2, \\ \lambda_1 &= \lambda_0 + p_a |g_{aw}|^2. \end{aligned} \quad (7)$$

Given N independent observations, the likelihood function for the observation vector $\mathbf{y}_W = [y_W[1], \dots, y_W[N]]^T$ is formulated as

$$p_i(\mathbf{y}_W) = \frac{1}{(\pi \lambda_i)^N} \exp\left(-\frac{\|\mathbf{y}_W\|^2}{\lambda_i}\right), \quad i \in \{0, 1\}. \quad (8)$$

For equiprobable hypotheses, the detection error probability (DEP) at Willie is defined as [8]

$$\xi = P_{MD} + P_{FA} = \Pr(\mathcal{D}_0 | \mathcal{H}_1) + \Pr(\mathcal{D}_1 | \mathcal{H}_0). \quad (9)$$

To minimize ξ , Willie employs an optimal likelihood-ratio detector. Based on (8), the log-likelihood ratio $\ln\left(\frac{p_1(\mathbf{y}_W)}{p_0(\mathbf{y}_W)}\right)$ relies strictly on the average energy of $\mathbf{y}_W$. Consequently, the optimal test simplifies to a radiometer detector: $T_W \stackrel{\mathcal{D}_1}{\geq} \varsigma$, where $T_W = \frac{1}{N} \sum_{n=1}^N |y_W[n]|^2$ is the test statistic and ς is the detection threshold. The minimum achievable DEP is lower-bounded by the total variation (TV) distance $V(p_0, p_1)$ as $\xi^* \geq 1 - V(p_0, p_1)$. Applying Pinsker's inequality upper-bounds the TV distance via the Kullback-Leibler (KL) divergence [13], yielding $\xi^* \geq 1 - \sqrt{\frac{1}{2} D(p_0 \| p_1)}$. Thus, satisfying the strict covertness requirement $\xi \geq 1 - \epsilon$ necessitates $D(p_0 \| p_1) \leq 2\epsilon^2$. By substituting (8) into the KL divergence definition $D(p_0 \| p_1) = \mathbb{E}_{p_0}[\ln \frac{p_0(\mathbf{y}_W)}{p_1(\mathbf{y}_W)}]$, we obtain a tractable sufficient covertness condition over N symbols:

$$D(p_0 \| p_1) = N \left(\ln \frac{\lambda_1}{\lambda_0} + \frac{\lambda_0}{\lambda_1} - 1 \right) \leq 2\epsilon^2. \quad (10)$$

III. PROBLEM FORMULATION AND OPTIMIZATION

A. Problem Formulation

In this work, we focus on the target tracking stage [1], [2], [3], [6], [7], treating the angle of arrival θ_0 as a priori information acquired during the initial detection phase.¹ Because the radar detection probability depends directly on the SINR [1], [3], we design the receive beamformer $\mathbf{u} \in \mathbb{C}^{N_r \times 1}$ to maximize the radar SINR, formulated from (6) as:

$$\eta^s = \frac{|\beta_0|^2 \mathbf{u}^H \mathbf{A}(\theta_0) \mathbf{R}_{xx} \mathbf{A}(\theta_0)^H \mathbf{u}}{\mathbf{u}^H (p_r \mathbf{h}_{rb} \mathbf{h}_{rb}^H + p_a \mathbf{h}_{ab} \mathbf{h}_{ab}^H + \mathbf{B} + \sigma_b^2 \mathbf{I}_{N_r}) \mathbf{u}}, \quad (11)$$

where $\mathbf{A}(\theta_0) = \mathbf{a}_r(\theta_0) \mathbf{a}_t^H(\theta_0)$, and $\mathbf{B} \triangleq \mathbf{R}_c + \beta \mathbf{H}_{SI} \mathbf{R}_{xx} \mathbf{H}_{SI}^H$ denotes the aggregate interference covariance.

In parallel, linear beamformers $\{\mathbf{w}_r, \mathbf{w}_a\}$ are employed for UL signal recovery. By introducing a generalized self-interference (SI) channel $\mathbf{C} \triangleq \beta_0 \mathbf{A}(\theta_0) + \sqrt{\beta} \mathbf{H}_{SI}$ that consolidates both SI and radar echoes, the SINRs for Roy and Alice are compactly formulated as

$$\eta^r = \frac{p_r \mathbf{w}_r^H \mathbf{h}_{rb} \mathbf{h}_{rb}^H \mathbf{w}_r}{\mathbf{w}_r^H (\mathbf{C} \mathbf{R}_{xx} \mathbf{C}^H + p_a \mathbf{h}_{ab} \mathbf{h}_{ab}^H + \mathbf{R}_c + \sigma_b^2 \mathbf{I}_{N_r}) \mathbf{w}_r}, \quad (12)$$

$$\eta^a = \frac{p_a \mathbf{w}_a^H \mathbf{h}_{ab} \mathbf{h}_{ab}^H \mathbf{w}_a}{\mathbf{w}_a^H (\mathbf{C} \mathbf{R}_{xx} \mathbf{C}^H + p_r \mathbf{h}_{rb} \mathbf{h}_{rb}^H + \mathbf{R}_c + \sigma_b^2 \mathbf{I}_{N_r}) \mathbf{w}_a}. \quad (13)$$

The achievable secrecy rate is given by $SR = [R_c - R_e]^+$, with transmission rates $R_k = \log_2(1 + \Upsilon_k)$ for $k \in \{c, e\}$. Let $\mathcal{A} \triangleq \{\{\mathbf{w}_r, \mathbf{w}_a, \mathbf{u}\}, \mathbf{v}_c, \mathbf{V}_0 \succeq \mathbf{0}, \{p_r, p_a \geq 0\}\}$ denote the optimization variable set. The joint optimization problem is formulated as follows:

$$\underset{\mathcal{A}}{\text{maximize}} \quad \log_2(1 + \eta^a) + SR \quad (14a)$$

$$\text{subject to} \quad \eta^s \geq \tau^{\text{rad}}, \eta^r \geq \tau^{\text{pub}}, \text{ and (10),} \quad (14b)$$

$$\|\mathbf{v}_c\|^2 + \text{Tr}(\mathbf{V}_0) \leq P^{\text{max}}, \quad (14c)$$

$$p_r \leq P^{\text{umax}}, p_a \leq P^{\text{umax}}. \quad (14d)$$

Here, τ^{rad} and τ^{pub} specify the minimum performance thresholds for radar sensing and Roy, respectively, while P^{max} and P^{umax} are the power budgets for FD-BS and UL users.

B. Sum Spectral Efficiency and Transmit Beamforming

We optimize the receive beamformers $\{\mathbf{w}_r, \mathbf{w}_a, \mathbf{u}\}$ by maximizing the SINR. As supported by established literature, the minimum mean square error (MMSE) receiver achieves the optimal solution [1], [2], [3]. Let us define the aggregate covariance matrices as $\Psi_s \triangleq p_r \mathbf{h}_{rb} \mathbf{h}_{rb}^H + p_a \mathbf{h}_{ab} \mathbf{h}_{ab}^H + \mathbf{B} + \sigma_b^2 \mathbf{I}_{N_r}$, $\Phi_r \triangleq p_a \mathbf{h}_{ab} \mathbf{h}_{ab}^H + \mathbf{R}_c + \mathbf{C} \mathbf{R}_{xx} \mathbf{C}^H + \sigma_b^2 \mathbf{I}_{N_r}$, and $\Phi_a \triangleq p_r \mathbf{h}_{rb} \mathbf{h}_{rb}^H + \mathbf{R}_c + \mathbf{C} \mathbf{R}_{xx} \mathbf{C}^H + \sigma_b^2 \mathbf{I}_{N_r}$. Because the noise term $\sigma_b^2 \mathbf{I}_{N_r}$ is full-rank, these matrices are strictly positive definite (i.e., $\Psi_s \succ \mathbf{0}$ and $\Phi_k \succ \mathbf{0}$ for $k \in \{r, a\}$). Consequently, the closed-form expressions are given by

$$\mathbf{u}^* = (\Psi_s)^{-1} \mathbf{a}_r(\theta_0), \mathbf{w}_k^* = (\Phi_k)^{-1} \mathbf{h}_{kb}, k \in \{r, a\}, \quad (15)$$

respectively. Substituting the optimal $\{\mathbf{u}^*, \mathbf{w}_r^*, \mathbf{w}_a^*\}$ into the SINR expressions in (11), (12) and (13) yields

$$\hat{\eta}^s = |\beta_0|^2 \mathbf{a}_t^H(\theta_0) \mathbf{R}_{xx} \mathbf{a}_t(\theta_0) \cdot \mathbf{a}_r^H(\theta_0) \Psi_s^{-1} \mathbf{a}_r(\theta_0), \quad (16a)$$

$$\hat{\eta}^r = p_r \mathbf{h}_{rb}^H \Phi_r^{-1} \mathbf{h}_{rb}, \hat{\eta}^a = p_a \mathbf{h}_{ab}^H \Phi_a^{-1} \mathbf{h}_{ab}. \quad (16b)$$

¹During this initial stage, the BS lacks prior target knowledge and transmits omni-directional signals or pilots to estimate target parameters.

Proof: Please refer to Appendix A. ■

Subsequently, applying (15), we rewrite (14) into the following tractable form with the reduced set $\tilde{\mathcal{A}} \triangleq \{\mathbf{v}_c, \mathbf{V}_0, p_r, p_a\}$:

$$\underset{\tilde{\mathcal{A}}}{\text{maximize}} \quad \log_2(1 + \hat{\eta}^a) + SR \quad (17a)$$

$$\text{subject to} \quad \hat{\eta}^s \geq \tau^{\text{rad}}, \hat{\eta}^r \geq \tau^{\text{pub}}, (10), (14c), (14d). \quad (17b)$$

To address the nonconvexity from the SINR constraints and objective function, we introduce auxiliary variables $\{u_a \geq 0, z_1, z_2, z_3, z_4\}$. Assuming Eve is an untrusted legitimate user, her CSI is available via standard pilots [2], [7], [16], [17]. Alternatively, for non-cooperative targets, we adopt a sensing-aided approach: the FD-BS reconstructs CSI via the LoS channel by estimating geometric parameters via extended Kalman filter (EKF) tracking states from radar echoes [14].

To address the non-convexity in the objective function, we rewrite the secrecy rate expression using a matrix-trace formulation. First, we define the following aggregate channel matrices: $\mathbf{H}_1 = [\sqrt{p_r} f_{rc}, \sqrt{p_a} f_{ac}] \in \mathbb{C}^{1 \times 2}$, $\mathbf{H}_2 = [\sqrt{p_r} u_{re}, \sqrt{p_a} u_{ae}] \in \mathbb{C}^{1 \times 2}$, $\mathbf{H}_3 = \mathbf{q}_{bc} \mathbf{q}_{bc}^H \in \mathbb{C}^{N_t \times N_t}$, $\mathbf{H}_4 = |\alpha_0|^2 \mathbf{a}_t(\theta_0) \mathbf{a}_t^H(\theta_0) \in \mathbb{C}^{N_t \times N_t}$. Using the identity $\mathbf{x}^H \mathbf{V} \mathbf{x} = \text{Tr}(\mathbf{V} \mathbf{x} \mathbf{x}^H)$, the quadratic terms in the SINR expressions can be converted into linear trace operations. The achievable rate terms $R_k = \log_2(1 + \Upsilon_k)$ for $k \in \{c, e\}$ are transformed using the relation $1 + \frac{S}{I+N} = \frac{S+I+N}{I+N}$. Consequently, the terms inside the logarithms for Carol and Eve are rewritten as

$$1 + \Upsilon_c = \frac{\text{Tr}(\mathbf{H}_1 \mathbf{H}_1^H) + \text{Tr}((\mathbf{V}_0 + \mathbf{V}_c) \mathbf{H}_3) + \sigma_c^2}{\text{Tr}(\mathbf{H}_1 \mathbf{H}_1^H) + \text{Tr}(\mathbf{V}_0 \mathbf{H}_3) + \sigma_c^2}, \quad (18)$$

$$1 + \Upsilon_e = \frac{\text{Tr}(\mathbf{H}_2 \mathbf{H}_2^H) + \text{Tr}((\mathbf{V}_0 + \mathbf{V}_c) \mathbf{H}_4) + \sigma_p^2}{\text{Tr}(\mathbf{H}_2 \mathbf{H}_2^H) + \text{Tr}(\mathbf{V}_0 \mathbf{H}_4) + \sigma_p^2}. \quad (19)$$

By imposing $R_c \geq R_e$, the secrecy rate reduces to $SR = R_c - R_e$, yielding the reformulated objective:

$$SR = \log_2 \left(\frac{\text{Tr}(\mathbf{H}_1 \mathbf{H}_1^H) + \text{Tr}((\mathbf{V}_0 + \mathbf{V}_c) \mathbf{H}_3) + \sigma_c^2}{\text{Tr}(\mathbf{H}_1 \mathbf{H}_1^H) + \text{Tr}(\mathbf{V}_0 \mathbf{H}_3) + \sigma_c^2} \right) - \log_2 \left(\frac{\text{Tr}(\mathbf{H}_2 \mathbf{H}_2^H) + \text{Tr}((\mathbf{V}_0 + \mathbf{V}_c) \mathbf{H}_4) + \sigma_p^2}{\text{Tr}(\mathbf{H}_2 \mathbf{H}_2^H) + \text{Tr}(\mathbf{V}_0 \mathbf{H}_4) + \sigma_p^2} \right). \quad (20)$$

To address the non-convex objective, we introduce auxiliary variables $\{z_k\}_{k=1}^4$. Defining the operator $\mathcal{T}(\mathbf{A}, \mathbf{B}, \mathbf{X}, n) \triangleq \text{trace}(\mathbf{A} \mathbf{A}^H) + \text{trace}(\mathbf{X} \mathbf{B}) + n$, the SR terms are reformulated as

$$e^{z_1} \leq \mathcal{T}(\mathbf{H}_1, \mathbf{H}_3, \mathbf{V}_0 + \mathbf{V}_c, \sigma_c^2), \quad (21a)$$

$$e^{z_2} \geq \mathcal{T}(\mathbf{H}_1, \mathbf{H}_3, \mathbf{V}_0, \sigma_c^2), \quad (21b)$$

$$e^{z_3} \geq \mathcal{T}(\mathbf{H}_2, \mathbf{H}_4, \mathbf{V}_0 + \mathbf{V}_c, \sigma_p^2), \quad (21c)$$

$$e^{z_4} \leq \mathcal{T}(\mathbf{H}_2, \mathbf{H}_4, \mathbf{V}_0, \sigma_p^2). \quad (21d)$$

Let $\mathcal{B} \triangleq \{\mathbf{V}_c, \mathbf{V}_0, \{z_k\}_{k=1}^4, p_r, p_a, u_a\}$ denote the set of optimization variables. The original problem (17) is recast as

$$\underset{\mathcal{B}}{\text{max}} \quad \log_2(1 + u_a) + \frac{1}{\ln 2} (z_1 - z_2 - z_3 + z_4) \quad (22a)$$

$$\text{s.t.} \quad z_1 - z_2 - z_3 + z_4 \geq 0, \quad (22b)$$

$$\mathbf{h}_{ab}^H \Phi_a^{-1} \mathbf{h}_{ab} \geq \frac{u_a}{p_a}, \quad \mathbf{h}_{rb}^H \Phi_r^{-1} \mathbf{h}_{rb} \geq \frac{\tau^{\text{pub}}}{p_r}, \quad (22c)$$

$$\mathbf{a}_r^H(\theta_0) \Psi_s^{-1} \mathbf{a}_r(\theta_0) \geq \frac{\tau^{\text{rad}}}{|\beta_0|^2 \mathbf{a}_t^H(\theta_0) \mathbf{R}_{xx} \mathbf{a}_t(\theta_0)}, \quad (22d)$$

$$\text{Tr}(\mathbf{V}_c) + \text{Tr}(\mathbf{V}_0) \leq P^{\text{max}}, \quad (22e)$$

$$\mathbf{V}_c, \mathbf{V}_0 \succeq \mathbf{0}, \quad p_r, p_a, u_a \geq 0, \quad (22f)$$

$$(10), (14d), (21). \quad (22g)$$

We drop the non-convex constraint $\text{rank}(\mathbf{V}_c) \leq 1$ via rank relaxation. To facilitate SCA, constraint (22d) is normalized by $\mathbf{a}_t^H(\theta_0)\mathbf{R}_{xx}\mathbf{a}_t(\theta_0)$. Subsequently, the convex term $\mathbf{a}_r^H(\theta_0)\Psi_s^{-1}\mathbf{a}_r(\theta_0)$ is lower-bounded by its first-order Taylor expansion around the previous iteration's solution $\Psi_s^{(i-1)}$ [2], [3]: $\mathbf{a}_r^H(\theta_0)\Psi_s^{-1}\mathbf{a}_r(\theta_0) \geq f_s(\mathbf{a}_r(\theta_0), \Psi_s, \Psi_s^{(i-1)})$, where $f_s(\cdot) = \mathbf{a}_r^H(\theta_0)(\Psi_s^{(i-1)})^{-1}(2\Psi_s^{(i-1)} - \Psi_s)(\Psi_s^{(i-1)})^{-1}\mathbf{a}_r(\theta_0)$. Identical affine approximations are employed to linearize $\mathbf{h}_{rb}^H\Phi_r^{-1}\mathbf{h}_{rb}$ and $\mathbf{h}_{ab}^H\Phi_a^{-1}\mathbf{h}_{ab}$. To address the fractional term $\frac{u_a}{p_a}$, we introduce an auxiliary variable t :

$$f_s(\mathbf{a}_r(\theta_0), \Psi_s, \Psi_s^{(i-1)}) \geq \frac{\tau^{\text{rad}}}{|\beta_0|^2}(\mathbf{a}_t^H(\theta_0)\mathbf{R}_{xx}\mathbf{a}_t(\theta_0))^{-1},$$

$$f_a(\mathbf{h}_{ab}, \Phi_a, \Phi_a^{(i-1)}) \geq t, \quad f_r(\mathbf{h}_{rb}, \Phi_r, \Phi_r^{(i-1)}) \geq \frac{\tau^{\text{pub}}}{p_r}. \quad (23)$$

Thus, bounding the objective auxiliary variable u_a necessitates the bilinear constraint $u_a \leq tp_a$. To guarantee a global concave lower bound for this term, we exploit the algebraic identity $tp_a = \frac{1}{4}[(t + p_a)^2 - (t - p_a)^2]$ and apply a first-order Taylor expansion to $(t + p_a)^2$ around $(t^{(i-1)}, p_a^{(i-1)})$, yielding

$$u_a \leq \frac{1}{2}(t^{(i-1)} + p_a^{(i-1)})(t + p_a) - \frac{1}{4}(t^{(i-1)} + p_a^{(i-1)})^2 - \frac{1}{4}(t - p_a)^2. \quad (24)$$

For the secrecy constraints, leveraging the monotonicity of the objective function, we approximate the non-convex terms in (21b) and (21c) via first-order Taylor expansions:

$$\mathcal{T}(\mathbf{H}_1, \mathbf{H}_3, \mathbf{V}_0, \sigma_c^2) \leq e^{z_2^{(i-1)}}(z_2 - z_2^{(i-1)} + 1), \quad (25a)$$

$$\mathcal{T}(\mathbf{H}_2, \mathbf{H}_4, \mathbf{V}_0 + \mathbf{V}_c, \sigma_p^2) \leq e^{z_3^{(i-1)}}(z_3 - z_3^{(i-1)} + 1). \quad (25b)$$

C. Covert Constraint With Imperfect CSI

We consider a practical scenario with imperfect CSI, characterized by norm-bounded errors for the UL channels and ellipsoidal uncertainty for the BS-Willie link, where $\mathbf{C}_{bw} \succ \mathbf{0}$ specifies the shape of the ellipsoid:

$$g_k = \hat{g}_k + \Delta g_k, \quad |\Delta g_k|^2 \leq v_k^2, \quad k \in \{aw, rw\}, \quad (26a)$$

$$\mathbf{g}_{bw} = \hat{\mathbf{g}}_{bw} + \Delta \mathbf{g}_{bw}, \quad \Delta \mathbf{g}_{bw}^H \mathbf{C}_{bw} \Delta \mathbf{g}_{bw} \leq o_{bw}^2. \quad (26b)$$

Leveraging the monotonicity of the KL divergence with respect to the ratio $\frac{\lambda_1}{\lambda_0}$ in (10), the covertness constraint is initially formulated as

$$p_a |g_{aw}|^2 + (1 - \kappa) \mathbf{g}_{bw}^H \mathbf{R}_{xx} \mathbf{g}_{bw} \leq (\kappa - 1) [p_r |g_{rw}|^2 + \sigma_w^2], \quad (27)$$

where the threshold $\kappa > 1$ satisfies $\ln \kappa + \frac{1}{\kappa} - 1 = \frac{2\epsilon^2}{N}$.

To strictly guarantee covertness under CSI uncertainty, we adopt a worst-case design. Applying the triangle inequality to maximize Alice's signal leakage while minimizing Roy's masking power, we decouple the norm-bounded UL errors, yielding the intermediate worst-case bound:

$$p_a (|\hat{g}_{aw}| + v_{aw})^2 \leq (\kappa - 1) [p_r (|\hat{g}_{rw}| + v_{rw})^2 + \mathbf{g}_{bw}^H \mathbf{R}_{xx} \mathbf{g}_{bw} + \sigma_w^2], \quad (28)$$

Algorithm 1: Proposed Algorithm to Solve (14).

- 1: **Initialize:** Iteration index $i = 0$, accuracy δ , and feasible initial points $\mathbf{V}_c^{(0)}, \mathbf{V}_0^{(0)}, \bar{p}_r^{(0)}, \bar{p}_a^{(0)}$.
- 2: **repeat**
- 3: **Solve** the convex sub-problem (30) to obtain optimal solutions $\{\bar{\mathbf{V}}_c, \bar{\mathbf{V}}_0, \bar{p}_r, \bar{p}_a\}$.
- 4: Update variables: $\mathbf{V}_c^{(i+1)} \leftarrow \bar{\mathbf{V}}_c, \mathbf{V}_0^{(i+1)} \leftarrow \bar{\mathbf{V}}_0$, etc.
- 5: Update iteration index: $i \leftarrow i + 1$.
- 6: **until** Convergence.
- 7: Construct the optimal beamforming vector $\mathbf{v}_c^*$ and AN covariance $\mathbf{V}_0^*$ and $p_k^* = \bar{p}_k$ for $k \in \{r, a\}$.
- 8: Compute receive beamformers $\{\mathbf{u}^*, \mathbf{w}_r^*, \mathbf{w}_a^*\}$ using (15).
- 9: **Output:** $\mathbf{v}_c^*, \mathbf{V}_0^*, p_r^*, p_a^*$ and receive beamformers.

where $(\cdot)_+ \triangleq \max(\cdot, 0)$. Next, we address the intractable semi-infinite constraint induced by the ellipsoidal uncertainty in $\mathbf{g}_{bw}$. By defining $\tilde{\mathbf{R}}_{xx} = (\kappa - 1)\mathbf{R}_{xx}$ and $q_1 = (\kappa - 1)[\sigma_w^2 + p_r(|\hat{g}_{rw}| + v_{rw})^2] - p_a(|\hat{g}_{aw}| + v_{aw})^2$, we exploit the S-procedure to recast (28) into a tractable linear matrix inequality (LMI). Introducing a slack variable $\eta_1 \geq 0$, this formulation guarantees robustness against any error realization within the ellipsoid:

$$\begin{bmatrix} \tilde{\mathbf{R}}_{xx} + \eta_1 \mathbf{C}_{bw} & \tilde{\mathbf{R}}_{xx} \hat{\mathbf{g}}_{bw} \\ \hat{\mathbf{g}}_{bw}^H \tilde{\mathbf{R}}_{xx} & \hat{\mathbf{g}}_{bw}^H \tilde{\mathbf{R}}_{xx} \hat{\mathbf{g}}_{bw} + q_1 - \eta_1 o_{bw}^2 \end{bmatrix} \succeq \mathbf{0}. \quad (29)$$

Proof: Please refer to Appendix B. ■

We hereby highlight the rationale for adopting the worst-case bound. In the context of covert communication, concealment is the paramount objective and serves as the absolute prerequisite for any information transfer, differing from rate-centric communications. Physically, the worst-case bound corresponds to the channel realization yielding the strongest signal at Willie. To strictly guarantee $\xi \geq 1 - \epsilon$, the system must be optimized against this scenario most favorable to detection, as relying on average CSI risks immediate exposure.

Finally, the original problem (14) is converted to the following problem at the i -th iteration:

$$\underset{\{B, t, \eta_1\}}{\text{maximize}} \quad \log_2(1 + u_a) + \frac{1}{\ln 2}(z_1 - z_2 - z_3 + z_4) \quad (30a)$$

$$\text{subject to} \quad (14d), (21a), (21d), (22e), (22f), (22b), (23), \\ (24), (25a), (25b), (29), t \geq 0, \eta_1 \geq 0. \quad (30b)$$

Problem (30) can be efficiently solved using CVX. By iteratively updating the local expansion points, the SCA algorithm yields a monotonically non-decreasing objective sequence within a bounded feasible set, ensuring convergence to a stationary point $\{\bar{\mathbf{V}}_c, \bar{\mathbf{V}}_0, \bar{p}_r, \bar{p}_a\}$. For practical implementation, a rank-one solution is readily extracted without performance loss [3]. Specifically, the optimal transmit beamformer is formulated as $\mathbf{v}_c^* = (\mathbf{q}_{bc}^H \bar{\mathbf{V}}_c \mathbf{q}_{bc})^{-1/2} \bar{\mathbf{V}}_c \mathbf{q}_{bc}$, enabling the exact recovery of the AN covariance via $\mathbf{V}_0^* = (\bar{\mathbf{V}}_c + \bar{\mathbf{V}}_0) - \mathbf{v}_c^* (\mathbf{v}_c^*)^H$. Together with the power allocations $p_k^* = \bar{p}_k$, the associated receive beamformers are computed using (15). The overall procedure is summarized in Algorithm 1.

Leveraging a pre-shared secret key and power codebook, the FD-BS embeds a covert grant within the downlink pilots. Alice extracts her power index via matching, public users process their pilots normally, while Willie, lacking the secret key, cannot reliably detect the grant from normal pilot distributions.

The interior-point method for solving (30) dictates an $\mathcal{O}(N_t^{6.5})$ complexity (marginally exceeding [2] due to (29)), making the SCA

TABLE II
SYSTEM CONFIGURATION PARAMETERS

Parameter	Value	Parameter	Value
Frequency (f_c)	6 GHz	PL (terrestrial user-BS)	-103.6 dB
Antennas (N_t, N_r)	(6, 6)	Radar Gain ($ \beta_0 ^2$)	-120 dB
BS Power ($P^{\max}$)	43 dBm	Rician Factor (K)	10 dB
UL Power ($P^{\max}$)	30 dBm	RSI Gain (β)	-110 dB
Noise Floor (σ_n^2)	-90 dBm	Target Angle (θ_0)	0 deg
Path Loss (PL) (UL-DL)	-110 dB	LoS Gain ($ \alpha_0 ^2$)	-90 dB
PL (Alice-Willie)	-90.5 dB	QoS Thresh. (τ^{pub})	8 dB
PL (Roy-Willie)	-98 dB	Radar Thresh. (τ^{rad})	15 dB
DEP Error ($\frac{2\epsilon^2}{N}$)	0.05	—	—

framework primarily a theoretical benchmark. To ensure scalability for large-scale arrays, a zero-forcing (ZF) heuristic is alternatively proposed. It decouples beam directions from power optimization by projecting the information and artificial noise beams into the null spaces of Eve's direction and Carol's channel, respectively. Moreover, since the monotonic detection constraint directly determines the covert power, a 2D search over the downlink power-splitting ratio and public uplink power on a grid G reduces the overall complexity to $\mathcal{O}(G^2(N_t^2 + N_r^3))$. Governed by $\mathcal{O}(N_t^2)$ transmit projections and $\mathcal{O}(N_r^3)$ MMSE matrix inversions per grid point, this approach relies on selecting a moderate G to balance search precision and computational cost.

IV. SIMULATION RESULTS

This section validates the proposed FD-ISAC system numerically. Key parameters are listed in Table II. We consider the following schemes across three performance domains:

- **Covert UL:** Compares the proposed joint cover (JC) scheme with baselines: public cover only (PCO), radar cover only (RCO), and no cover (NC), which represent scenarios lacking specific covering functionalities.
- **Secure DL:** Evaluates the proposed scheme (With AN) with its variants: random AN and without AN under perfect CSI. Specifically, to isolate the security gain yielded by our SE maximization strategy, we configure the proposed scheme to maximize the DL secrecy SE. For an energy-efficiency comparison, [2] is adapted to minimize DL power subject to the required UL covertness and QoS thresholds.
- **Sensing:** With benchmarks from secure analysis, we add the radar-only scheme to evaluate sensing performance.

Fig. 2 shows that covert SE generally degrades as channel uncertainty ρ increases. While the RCO scheme achieves the highest covert SE by avoiding public-UL interference, it exhibits greater sensitivity to CSI errors. In contrast, the proposed JC scheme demonstrates improved robustness with a slower decay rate, despite accommodating unavoidable public traffic. Furthermore, PCO outperforms the NC baseline but trails behind the radar-assisted schemes (JC/RCO), indicating that high-power radar signals play a dominant role in masking UL transmissions. Consequently, although RCO maximizes covert SE, only the JC scheme achieves a practical trade-off, balancing satisfactory covertness with guaranteed public QoS.

Fig. 3(a) quantifies the fundamental radar-communication trade-off inherent in ISAC: increasing the radar sensing threshold τ^{rad} from 9 to 17 dB reduces the achievable secrecy SE from 4.9 to 4.2 bps/Hz. Despite this decline, the integration of optimized AN demonstrably enhances secrecy performance, avoiding the sharper drops observed in baselines employing random AN or without AN. Moreover, our framework achieves a higher secrecy SE than power-min [2]. Beyond the inherent advantage of a rate-centric objective, this direct formulation

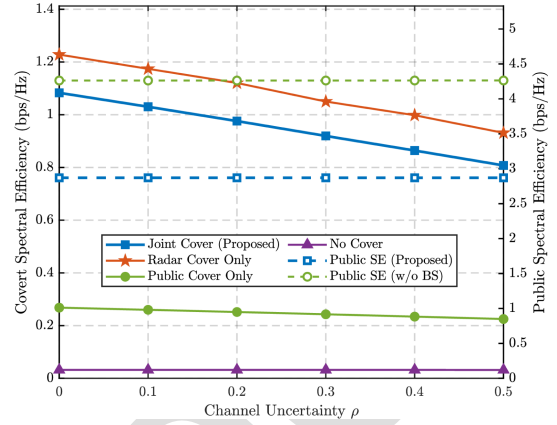


Fig. 2. Impact of normalized CSI errors on SE with $\rho = \sigma_{bw}^2 = v_{aw}^2 = v_{rw}^2$.

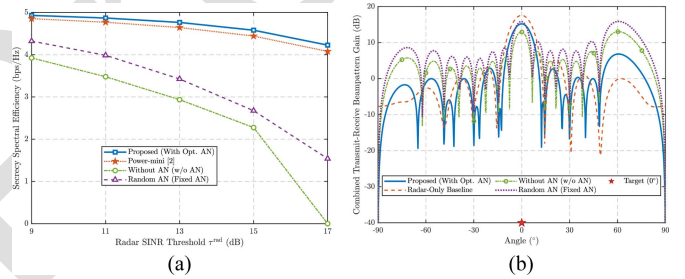


Fig. 3. (a) Secrecy performance versus τ^{rad} . (b) Radar sensing performance.

allows the algorithm to converge to a more rate-favorable stationary point within the non-convex feasible region.

Fig. 3(b) illustrates the combined transmit-receive beam pattern, formulated as $P_{\text{comb}}(\theta) = |\mathbf{u}^H \mathbf{a}_r(\theta)|^2 \mathbf{a}_t^H(\theta) \mathbf{R}_{xx} \mathbf{a}_t(\theta)$. By precisely directing a focused mainlobe toward the target (0°), the proposed method readily guarantees the mandated radar SINR threshold. Concurrently, it outperforms communication-centric baselines through substantial sidelobe suppression. This dual ability to maintain peak target illumination while minimizing unintended emissions underscores its robust interference mitigation.

V. CONCLUSION

This work proposes a bidirectional security framework for FD-ISAC, exploiting FD signals and AN for simultaneous UL covertness and DL secrecy. The resulting non-convex FD-coupled optimization problem is efficiently solved via SCA and rank relaxation. Extensive simulations validate the robustness and effectiveness of the proposed design. Future work will leverage UL signals to mask DL transmissions against successive interference cancellation (SIC)-capable eavesdroppers and integrate non-orthogonal multiple access (NOMA) for interference mitigation.

APPENDIX A

Recalling the radar SINR definition in (11), we have

$$\eta^s = \frac{|\beta_0|^2 (\mathbf{a}_t^H(\theta_0) \mathbf{R}_{xx} \mathbf{a}_t(\theta_0)) |\mathbf{u}^H \mathbf{a}_r(\theta_0)|^2}{\mathbf{u}^H \mathbf{\Psi}_s \mathbf{u}}. \quad (31)$$

By substituting the optimal MMSE receiver $\mathbf{u}^* = \Psi_s^{-1} \mathbf{a}_r(\theta_0)$ into (31), the optimal SINR is derived as

$$\begin{aligned} \hat{\eta}^s &= \frac{|\beta_0|^2 (\mathbf{a}_t^H(\theta_0) \mathbf{R}_{xx} \mathbf{a}_t(\theta_0)) |\mathbf{a}_r^H(\theta_0) \Psi_s^{-1} \mathbf{a}_r(\theta_0)|^2}{\mathbf{a}_r^H(\theta_0) \Psi_s^{-1} \Psi_s \Psi_s^{-1} \mathbf{a}_r(\theta_0)} \\ &= \frac{|\beta_0|^2 (\mathbf{a}_t^H(\theta_0) \mathbf{R}_{xx} \mathbf{a}_t(\theta_0)) (\mathbf{a}_r^H(\theta_0) \Psi_s^{-1} \mathbf{a}_r(\theta_0))^2}{\mathbf{a}_r^H(\theta_0) \Psi_s^{-1} \mathbf{a}_r(\theta_0)} \\ &= |\beta_0|^2 \underbrace{[\mathbf{a}_t^H(\theta_0) \mathbf{R}_{xx} \mathbf{a}_t(\theta_0)]}_{\text{Transmit Gain}} \underbrace{[\mathbf{a}_r^H(\theta_0) \Psi_s^{-1} \mathbf{a}_r(\theta_0)]}_{\text{Receive Gain}}. \end{aligned} \quad (32)$$

The derivations for $\hat{\eta}^r$ and $\hat{\eta}^a$ follow an identical procedure and are omitted for brevity. ■

APPENDIX B

To guarantee robustness against the ellipsoidal uncertainty defined by $\Delta \mathbf{g}_{bw}^H \mathbf{C}_{bw} \Delta \mathbf{g}_{bw} \leq o_{bw}^2$ with $\mathbf{C}_{bw} \succ \mathbf{0}$, we rewrite the worst-case constraint (28) in a quadratic form:

$$\begin{aligned} \mathcal{Q}(\Delta \mathbf{g}_{bw}) &\triangleq \Delta \mathbf{g}_{bw}^H \tilde{\mathbf{R}}_{xx} \Delta \mathbf{g}_{bw} \\ &+ 2\text{Re}\{\tilde{\mathbf{g}}_{bw}^H \tilde{\mathbf{R}}_{xx} \Delta \mathbf{g}_{bw}\} + C_0 \geq 0, \end{aligned} \quad (33)$$

where $C_0 = \hat{\mathbf{g}}_{bw}^H \tilde{\mathbf{R}}_{xx} \hat{\mathbf{g}}_{bw} + q_1$.

By virtue of the S-procedure, the implication $\Delta \mathbf{g}_{bw}^H \mathbf{C}_{bw} \Delta \mathbf{g}_{bw} \leq o_{bw}^2 \Rightarrow \mathcal{Q}(\Delta \mathbf{g}_{bw}) \geq 0$ holds if and only if there exists a slack variable $\eta_1 \geq 0$ such that

$$\mathcal{Q}(\Delta \mathbf{g}_{bw}) - \eta_1 (o_{bw}^2 - \Delta \mathbf{g}_{bw}^H \mathbf{C}_{bw} \Delta \mathbf{g}_{bw}) \geq 0, \forall \Delta \mathbf{g}_{bw}. \quad (34)$$

Rearranging the terms into a matrix structure yields the equivalent LMI:

$$\Xi = \begin{bmatrix} \tilde{\mathbf{R}}_{xx} + \eta_1 \mathbf{C}_{bw} & \tilde{\mathbf{R}}_{xx} \hat{\mathbf{g}}_{bw} \\ \hat{\mathbf{g}}_{bw}^H \tilde{\mathbf{R}}_{xx} & \hat{\mathbf{g}}_{bw}^H \tilde{\mathbf{R}}_{xx} \hat{\mathbf{g}}_{bw} + q_1 - \eta_1 o_{bw}^2 \end{bmatrix} \succeq \mathbf{0}. \quad (35)$$

Thus, (28) is satisfied provided $\Xi \succeq \mathbf{0}$ and $\eta_1 \geq 0$. ■

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